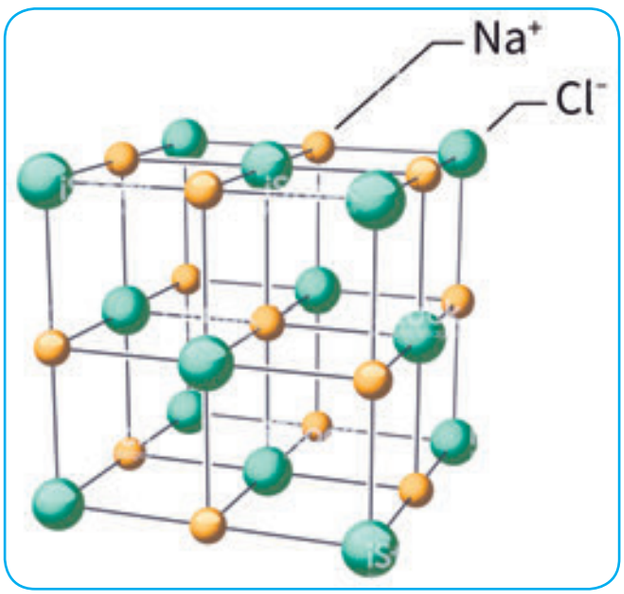
6.3 Classification of crystalline solids:

6.3.1 Ionic solids:

The structural units of an ionic crystal are cations and anions. They are bound together by strong electrostatic attractive forces. To maximize the attractive force, cations are surrounded by as many anions as possible and vice versa. Ionic crystals possess definite crystal structure; many solids are cubic close packed. Example: The arrangement of Na^+ and Cl^- ions in NaCl crystal.



Characteristics:

1. Ionic solids have high melting points.
2. These solids do not conduct electricity, because the ions are fixed in their lattice positions.
3. They do conduct electricity in molten state (or) when dissolved in water because, the ions are free to move in the molten state or solution.
4. They are hard as only strong external force can change the relative positions of ions.

6.3.2 Covalent solids:

In covalent solids, the constituents (atoms) are bound together in a three dimensional network entirely by covalent bonds. Examples: Diamond, silicon carbide etc. Such covalent network crystals are very hard, and have high melting point. They are usually poor thermal and electrical conductors.



6.3.3 Molecular solids:

In molecular solids, the constituents are neutral molecules. They are held together by weak van der Waals forces. Generally molecular solids are soft and they do not conduct electricity. These molecular solids are further classified into three types.



Graphite is used inside pencils.it slips easily off the pencil onto the paper and leaves a blackmark. Graphite is also a component of many lubricants , for example bicycle chain oil , because it is slippery

(i) Non-polar molecular solids:

In non polar molecular solids constituent molecules are held together by weak dispersion forces or London forces.